

Zack LaPrade

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EDUCATION

University of Georgia

Bachelor of Science, Computer Engineering

Scholarships: Zell Miller Scholarship

Athens, GA

May 2027

Cumulative GPA: 3.54/4.00

WORK AND RESEARCH EXPERIENCE

Technology Service Corporation — Defense Contractor

Electrical Engineering Intern

Huntsville, AL

May - August 2026

- Researched and tested an Over-Current Detection device, detecting over-current in devices within 300ns
- Created schematics and PCB layout for Over-Current Detection and several other schematics
- Debugged and validated units through measures such as X-Rays, probing voltages/resistances, and replacing components
- Gained experience using radio frequency and a Network Analyzer to scan the Quiet Zone of a compact range

UGA Robotics Research

Undergraduate Researcher

Athens, GA

February 2026 - Present

- Working with a team to develop a glove that would allow the user to control a robotic hand across the room
- Implementing the required sensors, including flex sensors and a 9-DOF IMU, to develop the project

Peer Learning Assistant

UGA Physics PLA

Athens, GA

August - December 2025

- Assisted the professor in answering questions/misconceptions from 100+ students in Physics 1 in lecture and/or lab
- Attended weekly peer education classes to practice effective communication methods when teaching and instructing

PROJECT EXPERIENCE

Raspberry Pi Smart Room System (MonCoeur)

May 2025 - Present

- Hosted a local server on a Raspberry Pi, logged sleep, hydration, exercise, and temperature to self-identify my migraine triggers and prevent them
- Utilized 4 API's, 20 threads, 21 methods, 3 sensors and 2 sockets to add room data, lightbulbs, and health data
- Achieved a hands-free and automatic experience through 4 different mediums: a Bluetooth button, voice commands, an iPhone app, and a wall display
- Applied offline speech recognition, sensor readings and API calls are controlled without internet or cloud service dependency

Multitool Bench

January - May 2026

- Worked with 4 classmates as hardware lead to replicate standard tools under the constraint of no IC's and weekly checkpoints
- Derived a stable +-12V supply through a class B push-pull amplifier with feedback to ensure no voltage sag
- Designed a Voltmeter and Ohmmeter using a 6-bit ADC and a Wheatstone bridge to negate noise with 2% error
- Created a frequency counter for any waveform utilizing a transistor with accuracy to 0.1 Hz

Smart Water Bottle

January - August 2026

- Integrated multiple sensors to measure water temperature, water level, and motion from the bottle for automatic logging and coaching on hydration
- Deployed HTTPS, TCP, and I2C to connect for sensor fusion and enable the software to send messages through NTFY
- Utilized Fusion 360 to design and 3D print a chassis to house the ESP32, sensors, and battery

MEMBERSHIPS AND ASSOCIATIONS

Theta Tau, Professional Engineering Fraternity

Rush / Recruitment Chair

Athens, GA

May 2025 - May 2026

- Collaborated with 2 co-chairs to plan and advertise all rush events and interviews for 250+ rushees
- Gained experience in public speaking, communication, and improved time management to effectively meet deadlines

IEEE, Institute of Electrical and Electronic Engineers, Member

August 2025 - Present

- Engaged in technical workshops and projects to enhance engineering skills, professionalism, and communication

TECHNICAL SKILLS AND LANGUAGES

Technical: Matlab, Altium, Fusion 360, Multisim, LTspice, Verilog, Assembly, Java, C, C++, Python, HTML, JavaScript, Parallel Programming, Circuit Analysis, PCB Design

Languages: English (Fluent), Spanish (Advanced), French (Advanced)

AWARDS AND ACCOLADES

International Baccalaureate Diploma, International Skills Diploma Seal, Georgia Seal of Biliteracy